

Evaluation and prediction of guide RNA activities in genome-editing tools

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Abstract

CRISPR genome-editing tools, including Cas9 and Cas12a nucleases, base editors and prime editors, have revolutionized genome manipulation across various species and cell types. These tools have undergone continuous improvement as new variants or types of editors have been generated to improve their efficiency, specificity and applicability. However, given the vast array of genome editors and the multitude of designable guide RNAs, selecting the optimal combinations for efficient and precise genome editing has become increasingly challenging, especially under variable experimental conditions. To address this issue, several methods for evaluating genome-editing tools in a high-throughput manner have been developed. The resulting large datasets of editing efficiencies or specificities have been used to develop machine learning models that predict efficiency and specificity, greatly facilitating the optimal selection of genome editors and guide RNAs. Here, we review recent developments in high-throughput evaluations and machine learning-based predictions of genome-editing efficiencies and/or off-target effects, together with recent advances in diverse genome-editing tools. We also cover artificial intelligence-based development and evolution of genome-editing tools.

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Key points

- Gene-editing tools such as CRISPR nucleases, base editors and prime editors can manipulate genomes across a wide range of species and cell types.
- With the expansion of the genome-editing toolbox, the selection of the most suitable tool and the design of appropriate guide RNAs for precise and efficient editing remain challenging.
- High-throughput methods enable the measurement of protospacer adjacent motif sequence preference, on-target and off-target editing efficiencies, and outcomes at diverse target sequences, facilitating the determination of optimal genome-editing tools and guide RNA combinations from the available options.
- The large amount of data on genome-editing tools evaluated at thousands of target DNA sequences has facilitated the development of machine learning models that quickly predict genome-editing activity for a given pair of guide RNA and target sequence.
- Large language models trained on DNA, RNA and protein data enable the artificial intelligence-based design and evolution of genome editors. Combined with high-throughput datasets, they can contribute to the development of next-generation genome-editing tools.

Introduction

Gene-editing tools such as CRISPR (clustered regularly interspaced short palindromic repeats) nucleases, base editors and prime editors have been successfully used to manipulate genomes across various species and cell types^{1–8}. These tools have enabled the development of animal models in mice⁹, dogs¹⁰, pigs¹¹ and monkeys¹², and have greatly expedited functional screening of genes and genetic variants^{13–24}. In 2023, a groundbreaking gene-editing agent targeting both sickle cell anaemia and β -thalassaemia²⁵ received FDA approval (NCT03655678, NCT03745287), demonstrating that a single round of genome-editing therapy can permanently treat a genetic disorder, and numerous other clinical trials are currently underway²⁶.

Gene-editing technology is rapidly advancing, expanding the range of available tools as researchers continue to explore and repurpose new CRISPR systems from bacteria^{27–34}. These efforts aim to address issues associated with the relatively large size of the prototypical Cas9 from *Streptococcus pyogenes* (SpCas9; ~4.1 kb) compared with smaller nucleases such as Cas9 from *Staphylococcus aureus* (SaCas9; ~3.2 kb) and *Campylobacter jejuni* (CjCas9; ~3.0 kb), the restrictions of protospacer adjacent motif (PAM) sequences and the potential for off-target editing. In 2016 and 2019, base and prime editors were developed by attaching deaminase^{35–37} or reverse transcriptase domains³⁸, respectively, to Cas9 nickases, enabling precise, efficient editing with minimal indels and double-strand break (DSB)-induced cytotoxicity. As the genome-editing toolbox expands, the tasks of selecting the appropriate gene-editing tool and designing optimal RNA components for the effective and precise introduction of desired edits have become increasingly challenging. The difficulty in selecting the appropriate tool lies in the variable efficiencies, specificities and compatibilities of the tools in the contexts of different desired edits and cell types. Additionally, designing the single guide RNAs (sgRNAs) and prime

editing guide RNAs (pegRNAs) that are loaded onto genome-editing proteins requires careful and detailed consideration for efficient and precise gene editing.

In this Review, we first explore the characteristics of widely used gene-editing tools, including CRISPR nucleases, base editors and prime editors. Next, we discuss the technical considerations of high-throughput evaluation techniques that measure on-target and off-target effects across numerous target sites for these tools. We then highlight the emerging synergy between large-scale datasets and advanced artificial intelligence (AI), focusing on machine learning models that predict sgRNA and pegRNA efficiencies and outcomes without wet-lab experiments, and introduce recent approaches for designing and evolving genome-editing tools using biological language models. Finally, we propose a perspective on the future directions of genome editing and high-throughput evaluation technologies.

Types of genome-editing tools

CRISPR nucleases

CRISPR nucleases, which consist of a Cas protein and an RNA component, are powerful tools capable of inducing DSBs at specific target DNA sequences. To date, two classes and six types of Cas genes have been identified³⁹ (Fig. 1a). Among these, the most widely used are class II, type II (for example, Cas9) and class II, type V (for example, Cas12a). In CRISPR–Cas9, the Cas9 effector protein and sgRNA, which is an engineered form of guide RNA derived from a naturally formed RNA duplex of CRISPR RNA (crRNA) and trans-activating crRNA, form a ribonucleoprotein complex (Fig. 1b). The Cas9 protein recognizes the PAM, a short DNA sequence adjacent to the target (protospacer) site; in the case of SpCas9, the PAM is NGG. The sgRNA guides the ribonucleoprotein to specifically bind the target DNA based on the complementarity between the guide sequence located at the 5' end of the sgRNA and the protospacer sequence in the target DNA.

Like previous generations of programmable nucleases such as zinc finger nucleases (ZFNs)^{40,41} and transcription activator-like effector nucleases (TALENs)⁴², CRISPR–Cas systems leverage the principle that DSBs substantially increase the probability of non-homologous end joining (NHEJ) or homology-directed repair (HDR)-mediated gene modifications⁴³. Indels induced by NHEJ can range from one base pair (bp) to several kilobases (kb) in length, resulting in frameshift-mediated gene knockouts. On one hand, the patterns and lengths of NHEJ-mediated indels are influenced by microhomology around the cleavage site and variations in the expression of cellular DNA repair pathways. On the other hand, HDR enables precise editing, replicating an exogenous donor DNA sequence to enable insertions, deletions or substitutions ranging from 1 bp to several kilobases in length. The efficiency of gene editing varies among cell types, with NHEJ-induced indels typically occurring at frequencies three to 20 times higher than HDR in mammalian cells^{3,44–46}. This difference results from the fact that HDR predominantly occurs during the G2 and late S phases of the cell cycle, whereas NHEJ is the dominant DSB-repair pathway across the cell cycle. Additionally, HDR requires an extra step in which the donor DNA template must be recruited and integrated at the DSB site. Therefore, these differences lead to an excess of indel generation via NHEJ, which represents a great challenge for achieving precise gene editing through HDR, especially in post-mitotic cells^{45–47}.

Cas9 nucleases derived from various bacteria continue to be engineered and repurposed for genome editing. SpCas9, the first to be harnessed for genome editing in human cells, has been engineered using directed evolution or rational design to reduce off-target editing

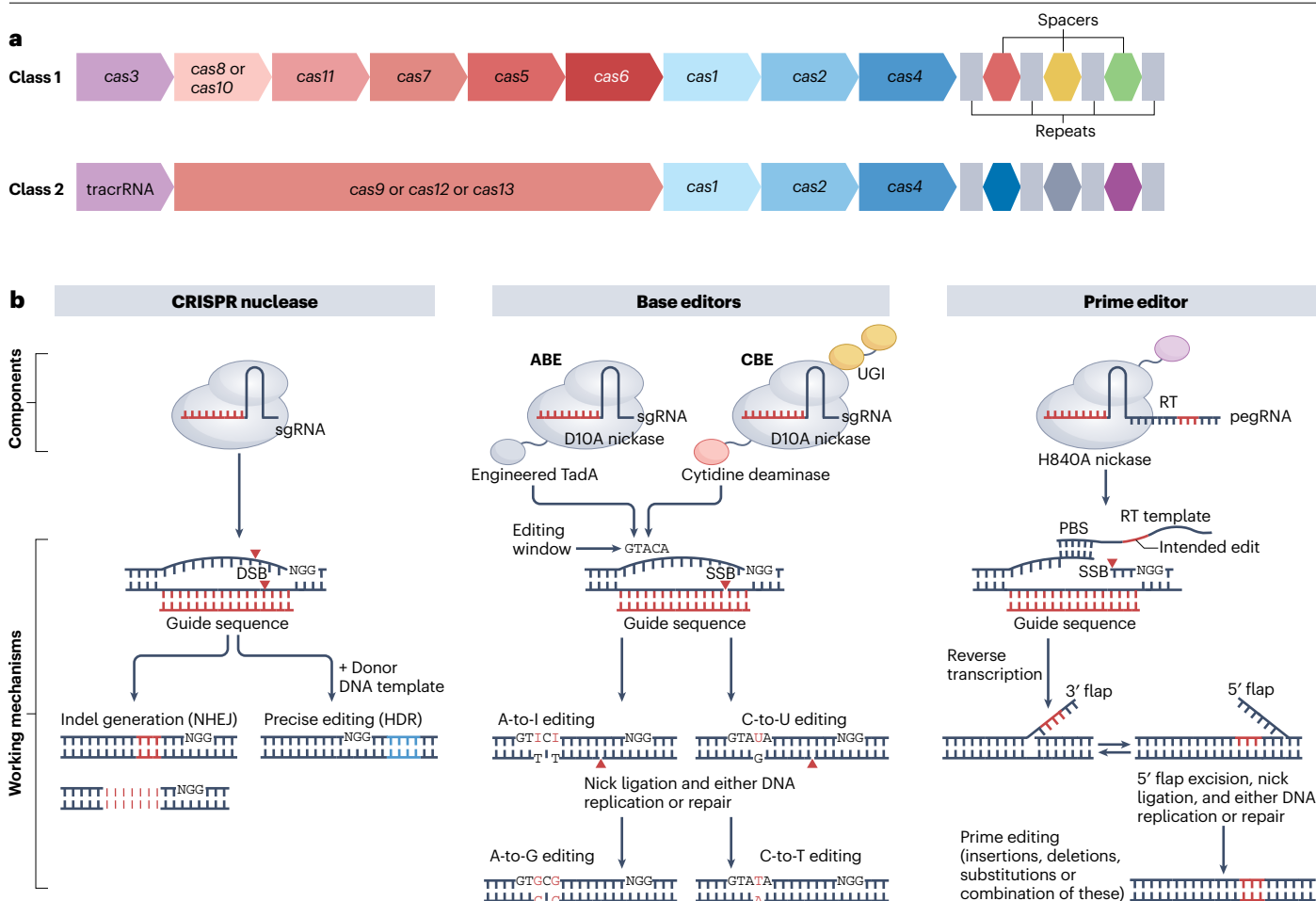


Fig. 1 | Genetic organization of the two Cas system classes and the mechanisms of genome-editing tools. a. Class 1 CRISPR–Cas systems are characterized by effector complexes made up of several distinct Cas proteins, whereas class 2 systems rely on a single, multidomain protein that interacts with CRISPR RNA. **b.** Components and working mechanisms of three types of genome-editing tools. CRISPR–Cas9 nuclease and single guide RNA (sgRNA) complex induce a double-stranded DNA break (DSB) at the target site. Non-homologous end joining (NHEJ) leads to indel generation, whereas homology-directed repair (HDR) enables precise editing in the presence of a donor template. Base editors consist of a Cas9 D10A nickase fused to a deaminase domain and induce cytosine

(C) or adenine (A) deamination on the non-target strand. Prime editors combine a Cas9 H840A nickase with a reverse transcriptase and use a prime editing guide RNA (pegRNA) that contains the guide sequence, primer-binding site (PBS) and reverse transcription template, enabling reverse transcription at the nicked site. For both base and prime editing, the modified sequence information on the non-target strand is copied to the target strand through cell replication or DNA repair. ABE, adenine base editor; Cas, CRISPR-associated; CBE, cytosine base editor; RT, reverse transcriptase domain; SSB, single-stranded DNA break; TadA, tRNA-specific adenosine deaminase; tracrRNA, trans-activating CRISPR RNA; UGI, uracil glycosylase inhibitor.

and/or broaden PAM compatibility. Cas9 variants, such as eSpCas⁴⁸, SpCas9-HF1⁴⁹, HypaCas9⁵⁰, xCas9⁵¹, evoCas9⁵² and SniperCas⁵³, have substantially improved editing specificity. When wild-type SpCas9 was the only choice, the requirement for an NGG PAM sequence restricted the scope of genome editing. The development of SpCas9 variants with altered PAM compatibilities, which include SpCas9-VRQR⁵⁴, SpCas9-NG⁵⁵, SpCas9-NRCH⁵⁶, SpCas9-NRRH⁵⁶, SpCas9-NRTH⁵⁶ and SpCas9-RY⁵⁷, has dramatically broadened the scope. Cas9 orthologues identified from other bacterial species have also been used in genome-editing applications. For instance, Cas9 orthologues that recognize PAM sequences other than NGG have been utilized to target DNA sequences lacking an NGG PAM. In addition, SaCas9²⁷ and CjCas9²⁸, with their smaller gene sizes (3.2 and 3.0 kb) relative to that of the SpCas9

gene (4.1 kb), are advantageous for recombinant adeno-associated virus vector production. Moreover, ultra-compact nucleases such as TnpB³⁸, Cas12f³² and LscB³³, with lengths around 1.2 to 2 kb, have been proposed for gene-editing applications. However, owing to their lower ‘on-target’ editing activity in their natural forms compared with SpCas9, various engineering strategies have been implemented to enhance their efficiency^{34,59–65}. For instance, protein and RNA engineering led to the development of elscB-D and eωRNA, which exhibit 20-fold increased activity compared with the wild-type LscB system⁶². Similarly, Cas12f activity has been considerably improved through deep mutational scanning and rational design, achieving efficiencies comparable with those of SpCas9 and Cas12a⁶³. In the case of TnpB, optimization of the ωRNA component has enhanced its gene-editing

Table 1 | Comparison of genome-editing tools

	Nucleases	Base editors	Prime editors
DNA breaks	DSB	SSB (target strand)	SSB (non-target strand) Two SSBs with variable offsets (for PE3, PE3b and PE5)
DNA repair system	NHEJ HDR	Base excision repair	Mismatch repair
RNA component	Single guide RNA	Single guide RNA	pegRNA (Optional) single guide RNA for PE3/PE5 approaches
Edit types	NHEJ-mediated insertions and deletions (indels) HDR-mediated precise insertions, deletions and substitutions	Base conversions (C to T (CBE), A to G (ABE), C to G (C to G base editor))	All possible mutations (insertions, deletions, and substitutions)
Edit lengths	NHEJ: typically less than 20bp HDR: 1bp to several kb	Single or multiple nucleotide conversion(s) within the editing window	~1–60bp when using a single pegRNA 1bp to 10kb when using pairs of pegRNAs
General efficiency	NHEJ: frequently very high (for example, 30% to 70%, sometimes more than 70%) HDR: generally low (~1–20%) and variable according to edit length and cell type	High and could be very high (for example, ~60% precise editing in mouse liver when the single target nucleotide is located in the centre of the editing window)	Variable according to edit length and cell type Could be substantially high (for example, ~46% prime editing in mouse liver)
Off-target effects	Guide-dependent off-target effects Undesired large deletions at on-target sites High-fidelity Cas variants have been developed	Guide-dependent off-target effects Guide-independent off-target effects in DNA and/or RNA High-fidelity deaminase domains have been developed	Almost undetectable when the number of mismatches in the region corresponding to the primer-binding site is more than three

ABE, adenine base editor; CBE, cytosine base editor; DSB, double-strand break; HDR, homology-directed recombination; NHEJ, Non-homologous end joining; PE, prime editor; pegRNA, prime editing guide RNA; SSB, single-strand break.

activity, enabling genome editing in cultured cells and phenotypic correction in animal models⁶⁴. These ultra-compact nucleases are particularly advantageous for delivery using adeno-associated virus (which has a limited cargo capacity) or lipid nanoparticles (Table 1).

RNA-targeting CRISPR nucleases, such as type VI (for example, Cas13a^{66,67}, Cas13b^{30,68} and Cas13d⁶⁹) and type III-E (such as Cas7-11^{70,71}) systems, have been developed for gene knockdown in diverse cell types, without introducing irreversible DNA alterations to treat viral infections^{72,73} and perform large-scale functional screenings of non-coding RNAs^{74–76}. Compared with conventional approaches such as antisense oligonucleotides and small interfering RNA, which often suffer from off-target effects, CRISPR systems have higher efficiency and specificity in RNA knockdown, highlighting their potential for more precise therapeutic applications.

Cas13 exhibits a unique characteristic known as collateral activity, which is linked to its structural organization^{67–69}. Cas13 becomes catalytically active only after forming a ternary complex with crRNA and complementary target RNA. The two catalytic domains of Cas13, higher prokaryotic and eukaryotic nucleotide-binding (HEPN) domains, cleave regions outside the crRNA–target RNA duplex and remain active even after the initial cleavage of the target RNA. This activity enables Cas13 to access and cleave non-specific RNA until the crRNA–target RNA duplex dissociates, leading to cell toxicity and growth inhibition in bacterial systems. Although collateral activity has been harnessed to develop highly sensitive and specific diagnostic tools^{77–79} by amplifying signals through bystander cleavage of probe RNA, there have also been attempts to enable RNA knockdown without collateral activity. For instance, engineered Cas13d variants⁸⁰ and Cas7-11^{70,71}, which exhibit minimal or absent collateral activity, have been applied for RNA knockdown. Engineered Cas13d variants achieve this by mutating the HEPN

domains responsible for non-specific RNA cleavage, thereby reducing collateral activity while retaining target specificity. By contrast, Cas7-11 lacks HEPN domains and uses Cas7-like domains to cleave target RNA within the guide–target duplex. These tools demonstrate lower toxicity and improved target specificity, making both Cas7-11 and engineered Cas13d promising candidates for therapeutic applications⁸¹.

Base editors

Base editors represent a class of DSB-free genome-editing tools, which overcome the issues of DSB-induced toxicity, although single nicks can still cause mild toxicity^{82,83} and an excess of unwanted indel generation. These editors use a Cas9 D10A nickase fused to a deaminase domain to induce specific base conversions. When a base-editor protein and sgRNA form a ribonucleoprotein complex and bind to a target, the target is unwound into two single strands, allowing the deaminase domain to catalyse the conversion of target nucleotides on the non-target single DNA strand (Fig. 1b).

The first-generation base editor (BE1) was developed by fusing a catalytically inactivated Cas9 with the rat cytidine deaminase rAPOBEC1, which converts cytosine to uracil within a small region (known as the base-editing window) in the target DNA. The effectiveness of base editing is influenced by DNA repair mechanisms such as base excision repair and mismatch repair (MMR). Deaminated nucleotides are recognized and processed by base excision repair pathways, potentially compromising the editing efficiency. To mitigate this problem, the second-generation base editor (BE2) includes a uracil glycosylase inhibitor domain to inhibit uracil DNA glycosylase, which initiates base excision repair by removing uracil from DNA. In the context of base editing, MMR machinery can recognize mismatches such as G–U, which result from C-to-U conversions induced by base editors.

Therefore, the third generation of base editor (BE3) incorporated D10A nickase to introduce a nick in the unedited DNA strand. This nick directs cells to preferentially use the U-containing edited strand as the correct template for DNA repair, thereby potentially increasing the frequency of intended C-to-T conversions.

Adenine base editors (ABEs) were developed by engineering the naturally occurring TadA to recognize and catalyse single-stranded DNA (ssDNA) substrates instead of RNA substrates. The TadA heterodimer, attached to the Cas9 D10A nickase, deaminates adenine in the non-target DNA strand to produce inosine. During DNA repair to resolve the T–I mismatch, inosine is converted into guanine, resulting in an A•T-to-G•C conversion. In the initial study, extensive directed evolution and protein engineering produced seventh-generation ABEs, which demonstrate high efficiency in converting A•T pairs to G•C pairs in human cells, with high purity characterized by minimal non-A-to-G conversion and indel generation.

The base-editing window of different base editors can vary depending on which Cas9 nickase and deaminase are used. For example, the windows of BE4 and ABE7.10 span positions 4 to 8, when nucleotides are counted from the end of the guide RNA-binding region that is distal to the PAM, whereas the window for Target-AID, which uses the PmCDA1 deaminase, spans positions 1 to 5. Generally, the efficiency of target nucleotide conversion is highest at the centre of the base-editing window and decreases towards the edges. Additionally, if multiple target nucleotides, such as adenines or cytosines, are present within the base-editing window, unwanted concurrent mutations (called bystander edits) might occur, in addition to the desired single-base conversion. To address this issue, base editors comprising different Cas9 variants, PAM compatibilities and deaminases, along with multiple sgRNAs, should be designed and evaluated to find the optimal combinations that achieve the highest editing efficiencies and purity.

Base editors can induce off-target effects in both a guide RNA-dependent and guide RNA-independent manner. Guide RNA-dependent off-target effects arise from guide RNA–DNA mismatches, similar to those induced by Cas9 nucleases. Although the deaminase domain is designed to operate only when the guide RNA binds to the target DNA, causing the non-target DNA strand to denature, and thus minimizing off-target effects, guide RNA-independent activity has nonetheless been observed for both ABEs⁸⁴ and cytosine base editors (CBEs)^{84–86}. Guide RNA-independent off-target effects occur primarily in transcriptionally active regions where the deaminase domain may interact with transiently exposed ssDNA substrates, or the residual RNA deaminase function of TadA can cause off-target effects in RNA transcripts. To address these issues, high-fidelity variants including ABE8e-V106W⁸⁷, ABE8.17-m+V106W⁸⁸, CBE-YE1-BE4max⁸⁹ and SsAPOBEC3B⁹⁰ have been developed via directed evolution or rational design of the deaminase domain. In 2023, the small yet highly active deoxyadenosine deaminase TadA was evolved into a cytidine deaminase, enabling the generation of base editors CBE-T⁹¹ and TadCBEs⁹², which exhibit similar or higher on-target activity and lower off-target activity compared with CBEs that use APOBEC deaminases. These advancements greatly reduce guide RNA-independent DNA and RNA off-target effects while maintaining or improving on-target activity, thereby alleviating safety concerns for therapeutic base editing.

RNA base editors are an emerging class of genome-editing tools that enable precise nucleotide conversions in RNA transcripts without altering the underlying genomic DNA sequence. These editors offer a new alternative to DNA base editors, particularly for applications requiring transient or reversible gene editing. Similar to DNA base

editors, RNA base editors use a catalytically inactivated RNA-targeting nuclease fused to a deaminase domain. The first RNA base editor, REPAIR³⁰, was developed by fusing dead Cas13b (dCas13b) with adenosine deaminase acting on RNA type 2 (ADAR2). This system catalyses the conversion of adenosine to inosine, which is recognized as guanosine by cellular machinery. In 2019, RESCUE⁹³ introduced cytosine deaminase activity into ADAR2 through directed evolution, enabling cytidine-to-uridine editing in RNA. Subsequent advancements have focused on improving the efficiency and specificity of RNA base editors, including engineering ADAR2 to minimize off-target effects⁹³ and incorporating new Cas13 orthologues to enhance editing efficiency^{94,95}. These developments have substantially expanded the capabilities of RNA base editing and facilitated its application across diverse cell types.

Prime editors

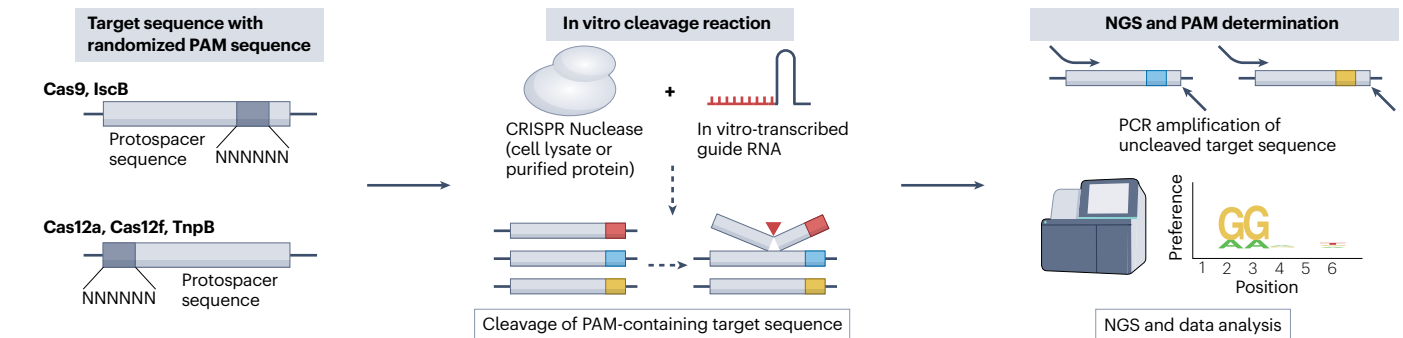
Prime editors combine the capabilities of a Cas9 nickase with a reverse transcriptase domain³⁸. Unlike base editors, which are limited to specific nucleotide conversions, prime editors can introduce any desired nucleotide change, as well as small insertions or deletions and combinations of these alterations, without inducing DSBs or requiring donor DNA templates (Fig. 1b).

In prime editing, the prime editor protein, composed of a Cas9 H840A nickase and an reverse transcriptase domain, forms a ribonucleoprotein complex with a pegRNA. Not only does the pegRNA guide the prime editor to the target DNA sequence, but it also, at its 3' end, contains a primer-binding site (PBS) and reverse transcription template (RTT) sequence for introducing the desired edit. The Cas9 H840A nickase induces a nick in the strand of the target DNA containing the PAM; subsequently, the 3' end of the exposed pegRNA binds to the PBS region of the protospacer sequence to initiate reverse transcription. The reverse transcriptase domain synthesizes the complementary DNA containing the desired edits and the target DNA homologous region. These steps create a branched intermediate with two redundant ssDNA flaps: a 3' flap that contains the edited sequence copied from the pegRNA and a 5' flap that contains the unedited DNA sequence. The 3' flap intermediate is thermodynamically favoured whereas 5' flaps are substrates for structure-specific endonucleases like FEN1, which excises 5' flaps. This preferential 5' flap excision helps incorporate the edited DNA strand, creating heteroduplex DNA with one edited and one unedited strand. Finally, DNA repair mechanisms resolve the heteroduplex by copying the information in the edited strand to the complementary strand, permanently installing the edit.

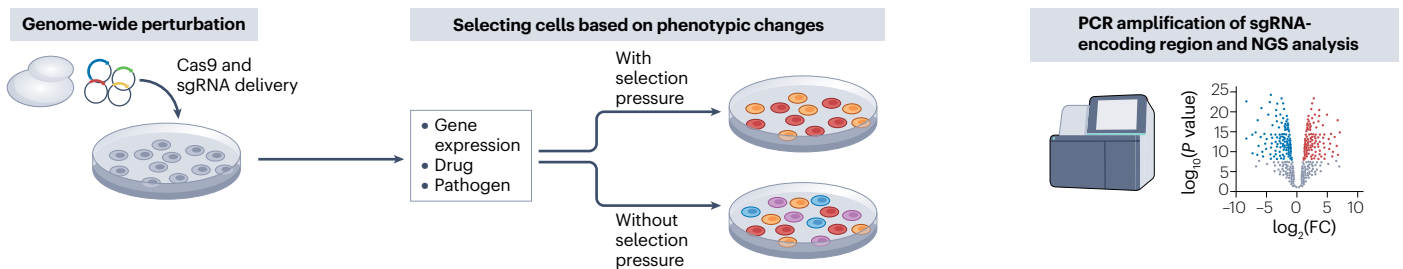
Prime-editing technology has been rapidly evolving, leading to enhanced efficiency and accuracy. The first-generation prime editor (PE1) was constructed by fusing the Cas9 H840A nickase with the reverse transcriptase from Moloney murine leukaemia virus (M-MLV RT). PE2 was created next by introducing a pentamutant M-MLV RT (D200N, L603W, T330P, T306K and W313F) to improve binding affinity, enzyme processivity and thermostability. PE3 was developed by adding an sgRNA designed to direct the introduction of a nick in the unedited DNA strand, increasing prime-editing efficiency. In 2021, PEmax was developed; it featured optimized codon usage, R221K and N394K mutations in the Cas9 domain, an additional nuclear localization signal at C-terminus of the Cas9 domain, and an optimized peptide linker⁹⁶. Moreover, Repair-seq identified DNA repair pathways influencing prime-editing efficiency, resulting in the PE4 and PE5 editing systems. In these systems, expression of dominant negative MLH1 was used to inhibit the MMR pathway, substantially improving prime editing⁹⁶.

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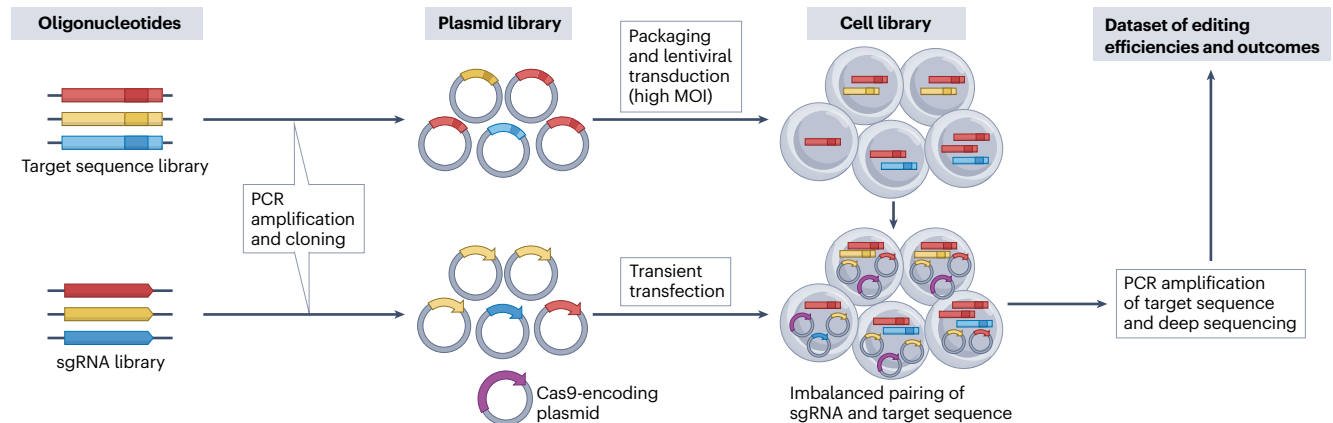
a In vitro cleavage assay (mainly for PAM determination)



b Profiling phenotypic changes: positive and negative selection



c Library-on-library



d Pairwise library

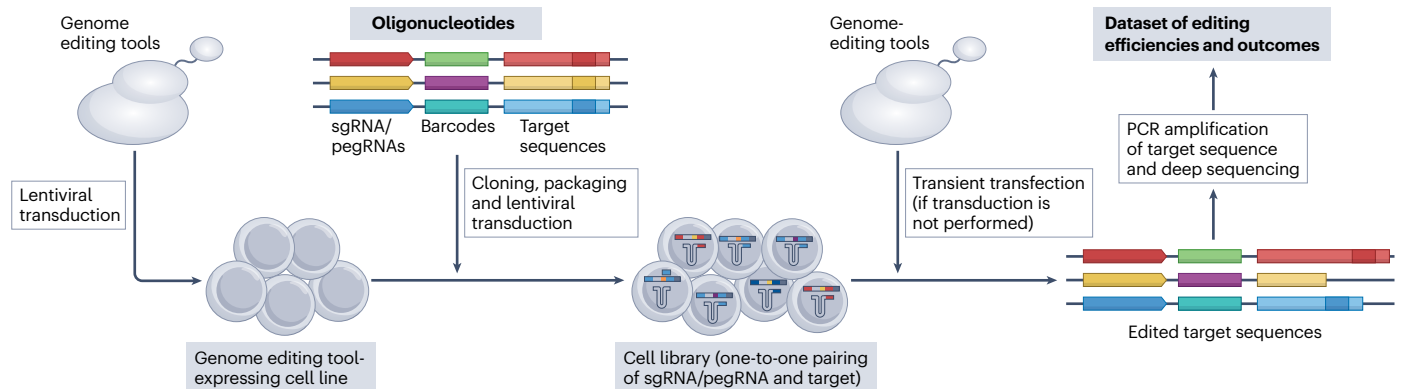


Fig. 2 | High-throughput evaluation methods for genome-editing tools.

a, Schematic representation of in vitro cleavage assay. **b**, Positive and negative selection based on phenotypic changes post genome-wide perturbation. **c**, Library-on-library approaches that deliver the target library and sgRNA library separately. **d**, Overview of the pairwise library method. By including the sgRNA or prime editing guide RNA (pegRNA) sequence with its corresponding target sequence in a single oligonucleotide, the issue of imbalanced pairing between the

sgRNA/pegRNA and the target in cells is overcome. Barcode sequences are used to determine the original form of the edited target sequence. A representative volcano plot is shown for illustrative purposes. FC, fold change in sgRNA read counts (with versus without selection pressure); MOI, multiplicity of infection; NGS, next-generation sequencing; PAM, protospacer adjacent motif; sgRNA, single guide RNA.

Compact prime editors, PE6a through PE6g, have also been developed by harnessing evolved Cas9 domains and small reverse transcriptase variants⁹⁷. These compact prime editors exhibit improved editing efficiency, and their reduced size makes them more suitable for dual adeno-associated virus-mediated in vivo delivery than PEmax. In 2024, incorporation of the small RNA-binding exonuclease protection factor La at the prime editor C terminus (PE7) further improved editing efficiency⁹⁸. For in vivo prime editing, split prime editors were developed and adopted to address the size limitations of adeno-associated virus packaging^{99–102}. Parallel to the engineering of prime-editor proteins, the addition of protective motifs at the 3' end of pegRNAs further enhances RNA stability and improves prime-editing efficiency¹⁰³.

Prime editors benefit from high editing purity and precision with lower risk of large genomic rearrangements and other unintended modifications associated with NHEJ. The prime-editing process involves multiple steps: Cas9-induced nick formation, reverse transcription and 3' flap resolution. The reason for the relatively low off-target effects of prime editing could be attributable to the low probability of all these conditions being met at off-target genomic locations^{104–106}.

High-throughput methods for evaluating genome-editing tools

The development of various genome-editing tools, including natural CRISPR systems and their derivatives, such as base and prime editors, has greatly expanded the capabilities and effectiveness of genome editing. However, this expansion has also made it more challenging to select the most suitable editing tool and to design sgRNAs or pegRNAs for accurate and efficient editing. As a result, there is an increasing need for techniques that can profile the characteristics of these genome-editing tools in a high-throughput manner.

In vitro cleavage assay

The in vitro cleavage assay is a quick and straightforward method for determining the PAM and/or guide RNA scaffold sequences recognized by a given CRISPR nuclease or for confirming the nuclease activity of novel CRISPR nuclease and sgRNA complexes^{27,29,107}. For PAM sequence determination, a DNA library containing regions corresponding to the guide sequence and random PAM sequences (multiple N-mers) is prepared and incubated with the Cas-sgRNA ribonucleoprotein complex (Fig. 2a). This assay can be conducted using Cas proteins either expressed and purified from bacteria or obtained from lysates of human cells transfected with Cas-encoding plasmids. For guide RNAs, either in vitro-transcribed or chemically synthesized sgRNAs are used. After incubation, target DNAs with the appropriate PAM sequences are cleaved by the Cas-sgRNA complex, whereas those without suitable PAM sequences remain uncleaved. The PAM sequence preferences of Cas proteins can then be determined by quantifying the depletion of target DNAs with specific PAM sequences using next-generation sequencing. In 2020, the high-throughput PAM determination assay (HT-PAMDA) was developed to evaluate hundreds of nucleases in a

multiplexed manner^{57,108}. HT-PAMDA uses fluorescent proteins to normalize the concentration of CRISPR nucleases used in the experiment, ensuring high reproducibility and enabling accurate comparisons. Additionally, it eliminates the need for protein purification by using human cell lysates. Importantly, this method was extended to profile base-editor PAM sequences; CBE-HT-PAMDA leverages USER enzyme treatment to excise uracil residues generated by C-to-U deamination, resulting in DSBs. Similarly, ABE-HT-PAMDA harnesses endonuclease V to cleave inosine-containing strands, enabling the detection of A-to-I deamination events.

Profiling phenotypic changes in target cells

High-throughput methods based on phenotypic changes provide a robust approach for evaluating the activity and specificity of genome-editing tools, especially CRISPR nucleases, by linking editing events to observable cellular outcomes. Cas9 can easily be programmed by modifying the 20-nucleotide guide sequence, which enables the construction of sgRNA libraries for genome-wide perturbation screening in a single experiment^{13–16}. Array-synthesized oligonucleotides are PCR-amplified and cloned into a lentiviral vector to generate an sgRNA-encoding plasmid library. When lentivirally delivered to Cas9-expressing cells, this library induces the knockout of target genes, leading to observable phenotypic changes (Fig. 2b).

In 2014, this method was first used for the high-throughput evaluation of SpCas9 activity in human and mouse cells¹⁰⁹. Cas9-expressing cell lines were infected with a lentiviral library containing 1,841 sgRNAs targeting human or mouse cell-surface markers at a low multiplicity of infection (<0.3), ensuring that each cell received a single sgRNA-encoding lentivirus. The frameshifts induced by Cas9 and sgRNAs led to the knockout of surface marker genes, and the marker-negative cell populations were isolated using FACS (fluorescence-activated cell sorting). The sgRNA-encoding sequences in these populations were then analysed using next-generation sequencing, revealing that sgRNAs enriched in the marker-negative cells exhibited higher activity than the marker-positive cells. Similarly, delivering sgRNAs targeting essential genes can disrupt cell proliferation, thereby enabling large-scale evaluation of Cas9 activity^{13–16,110}. In addition to approaches inducing cell dormancy or depletion, positive selection strategies, in which specific gene knockouts enable cell survival, have also been used to measure the efficiency of genome-editing tools. For example, cells treated with 6-thioguanine^{13,14} or a therapeutic RAF inhibitor¹⁵ can survive if *HPRT1*, MMR pathway genes or drug resistance-associated genes are knocked out. Sequencing of sgRNA-encoding regions from surviving cells enables the identification of highly active sgRNAs. These methods are versatile and can be used not only for measuring Cas9 activity but also for determining PAM sequences and assessing off-target effects in both human and bacterial cells. For instance, phenotypic change-based approaches have been adopted to evaluate the activity of SpCas9 using approximately 70,000 sgRNAs¹¹¹ and to determine the PAM sequence of SpCas9

variants⁵⁴ and SaCas9 (ref. 112) in *Escherichia coli*. The off-target effects of SpCas9 have also been examined using a library of matched and mismatched sgRNAs targeting surface markers¹¹⁰ in mammalian cells.

However, these methods have limitations in directly measuring genome editing at the DNA level, which complicates the assessment of modifications that do not result in clear phenotypic changes. For example, in-frame indels caused by Cas9, as well as base and prime edits that do not lead to a functional knockout, are particularly challenging to evaluate with these approaches. Furthermore, the choice of cell line imposes additional constraints; when targeting surface markers, only cell lines that express the specific genes can be used, which restricts the ability to measure sgRNA activity across different experimental conditions. Additionally, variations in the extent of editing and the resulting phenotypic changes must be considered, as these factors can hinder the comparison of the activity of sgRNAs targeting different genes. Moreover, gene function can vary considerably depending on the cell type, adding another layer of complexity to the interpretation of results. Despite these challenges, high-throughput phenotypic screening methods remain indispensable for understanding the functional aspects of genome-editing tools, including their efficiency and specificity in diverse experimental contexts.

Pairwise libraries

To evaluate genome-editing tools in a high-throughput manner, in 2015, methods were developed for simultaneously delivering sgRNAs and exogenous target DNA sequences into cells¹¹³. In this library-on-library approach, a target DNA-encoding lentiviral library is introduced into cells, followed by transfection with a corresponding sgRNA-encoding plasmid library. This approach allows for the measurement of indel frequencies across over 1,000 target DNA sequences simultaneously. However, imbalanced pairing between lentivirally delivered target DNA and plasmid-encoded sgRNA sequences complicates accurate indel frequency measurement (Fig. 2c). Additionally, introducing multiple copies of the target DNA and the sgRNA library can cause toxicity due to extensive DSBs.

This issue was addressed by using a pairwise library of guide RNA and target sequences^{114–117} (Fig. 2d). The pairwise library can be generated using oligonucleotides, such that each oligonucleotide contains both a guide RNA sequence and a corresponding target DNA sequence. This method ensures that each cell harbours an sgRNA and its corresponding target sequence, effectively overcoming such limitations. The pool of oligonucleotides is array-synthesized, PCR-amplified and assembled into a lentiviral vector. Next, lentiviral libraries are produced and delivered into cells at a low multiplicity of infection (< 0.3), ensuring that each cell contains a single copy of the guide sequence and corresponding target sequence pair. Upon delivery of genome-editing proteins by transient transfection or lentiviral transduction, indels are induced at the integrated target sequences based on the activities of the sgRNAs.

One of the key advantages of this technology is its flexibility, enabling guide sequences and target sequences to be specifically designed according to the desired purpose. The library can include matched or mismatched guide sequences paired with matched or mismatched target sequences, or target sequences with randomized PAM sequences. In 2017, this method was developed and adopted for high-throughput evaluation of editing efficiency, specificity and PAM sequence preferences of two Cas12a nucleases, one from *Acidaminococcus* sp. BV3L6 and the other from Lachnospiraceae bacterium ND2006, in human cells¹¹⁴.

The pairwise library approach enables high-throughput evaluation of editing efficiencies and outcomes directly in target sequences. The method has been refined and applied to a broad range of genome-editing tools, including SpCas9 and its variants^{115–120}, Cas9 orthologues¹²¹, and diverse types of base editors^{122–127} and prime editors^{128–132}. In addition, this approach has expanded the scope of evaluation by not only measuring the editing frequencies but also considering the editing patterns and outcomes in target sequences^{115,116,122,123,127}. The pairwise library approach has also been incorporated into the TRIP (thousands of reporters integrated in parallel) method¹³³ to determine the chromatin environment of the integrated target sites, enabling the measurement of the effect of chromatin context on prime-editing efficiency¹³². Moreover, tested on multiple cell lines, each expressing a distinct Cas9 nuclease or base editor, this method has facilitated comparison of PAM sequences, editing efficiencies and editing specificities of various Cas9 nucleases¹¹⁸, base editors¹²⁶ and prime editors¹²⁸. The large datasets generated from this approach have been used to develop machine learning models, which assist in the selection of genome-editing tools and the design of sgRNAs/pegRNAs (Fig. 3a). This approach has also been applied to optimize prime-editing-based therapeutics⁹⁹, for which numerous potential pegRNA options are often available for introducing a specific mutation (Fig. 3b). Furthermore, it has greatly improved the efficiency and precision of genome-wide functional screening with base¹⁸ and prime editors¹³⁴ (Fig. 3c and Box 1).

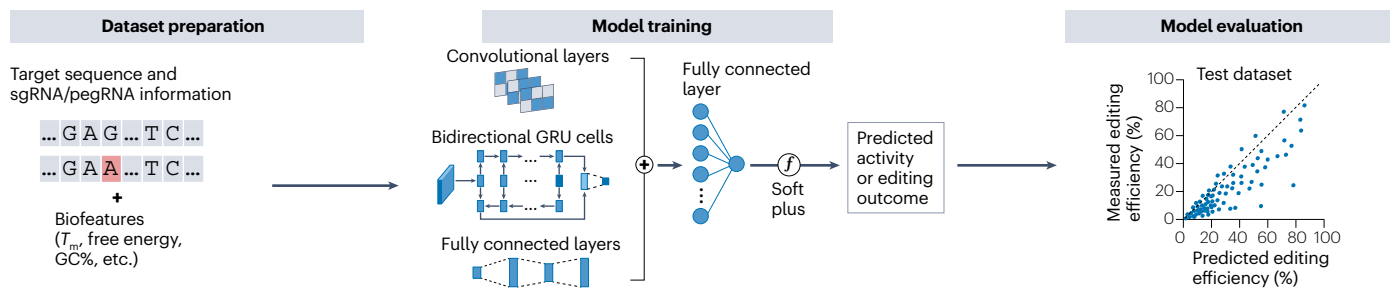
Despite these advantages, the pairwise library approach also has notable practical limitations. First, the decoupling of library members (for example, guide sequences and corresponding target sequences) may occur during PCR amplification of guide-encoding and target sequence-encoding regions¹³⁵, or during lentiviral integration of sequence pairs¹³⁶, driven by homologous regions between members, which could compromise the integrity of the results. Second, the substantial costs associated with oligonucleotide library synthesis, cell culture and deep sequencing make large-scale studies resource-intensive and operationally challenging. Finally, the outcomes of high-throughput experiments can vary depending on experimental parameters such as cell type, delivery method and incubation conditions, which hinders reproducibility and the generalization of results across laboratories.

Unbiased methods for detecting genome-wide off-target effects

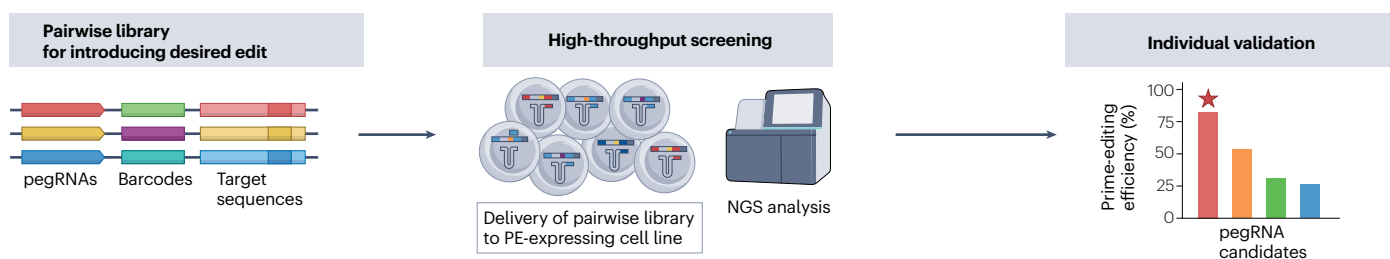
Although unbiased off-target detection methods such as digenome-seq¹³⁷, GUIDE-seq¹³⁸, CIRCLE-seq¹³⁹, DISCOVER-Seq¹⁴⁰ and CHANGE-seq¹⁴¹ (Fig. 4) were originally developed to identify unintended cleavage sites for specific sgRNA–Cas9 combinations, they can also be leveraged to compare the general specificity of genome-editing tools by systematically identifying potential off-target effects across the genome.

Digenome-seq identifies potential off-target sites by comparing the sequencing patterns of Cas9-treated genomic DNA with those in untreated control DNA. The approach uses *in vitro* digestion to detect cleavage sites with high sensitivity, enabling the identification of off-target sites where indels occur at frequencies as low as 0.1%¹³⁷. DIG-seq¹⁴² improves this approach by incorporating chromatin information, and multiplex digenome-seq enables simultaneous identification of off-target sites for multiple sgRNAs. By contrast, CIRCLE-seq and CHANGE-seq harness circular DNA libraries generated by ligation of linearized genomic DNA. In CIRCLE-seq, the library is then incubated with a Cas9–sgRNA complex, and circular DNAs containing potential

a Development of machine learning models



b Optimization of gene-editing therapeutics



c Genome-wide functional screening

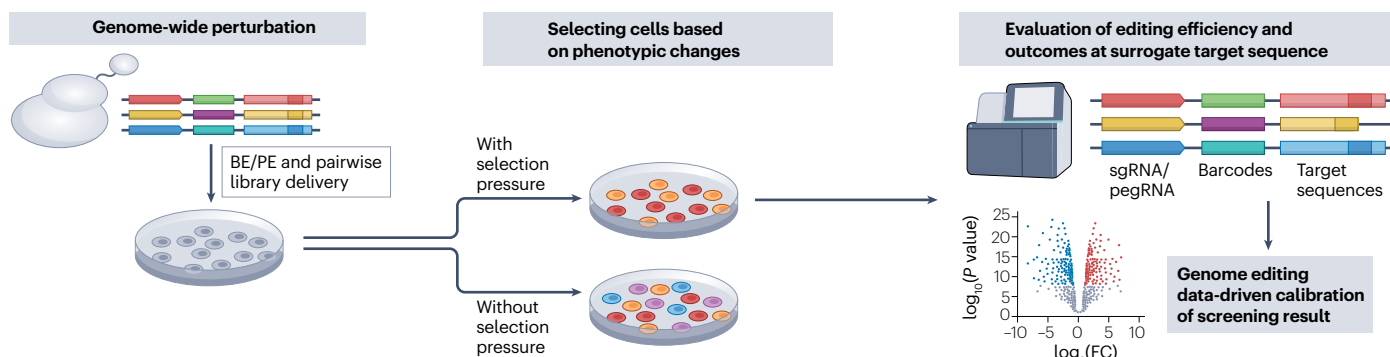


Fig. 3 | Applications of pairwise libraries of guide RNA and target sequences.

a, Development of machine learning models that predict the efficiencies and outcomes of genome-editing tools using datasets obtained from pairwise library-based experiments as the training data. Conventional machine learning and deep learning approaches are used depending on the size of the dataset. For model evaluation, datasets that were never used for training are utilized. **b**, Experimental selection of the most efficient and precise pegRNA for introducing an intended edit from thousands of candidates. **c**, Application of a pairwise library in genome-wide functional screening. By using editing

efficiencies and outcomes measured at surrogate target sequences, sgRNAs that efficiently produced a single predominant protein variant or pegRNAs with sufficiently high efficiency to induce phenotypic changes can be identified and used for functional evaluations. For more details, refer to Box 1. BE, base editor; FC, fold change in sgRNA read counts (with versus without selection pressure); GC, guanine-cytosine contents; GRU, gated recurrent unit; NGS, next-generation sequencing; PE, prime editor; pegRNA, prime editing guide RNA; sgRNA, single guide RNA; T_m , melting temperature. Part **a** adapted with permission from ref. 128, Elsevier.

off-target sites are cleaved; the cleaved DNA fragments are then prepared for deep sequencing. CHANGE-seq improves the compatibility and detection sensitivity by using Tn5 tagmentation for efficient circularization prior to the in vitro cleavage reaction.

GUIDE-seq enables the sensitive identification of off-target sites in live cells by integrating a double-stranded oligodeoxynucleotide tag into DNA breaks. The tagged sites can then be amplified using primers complementary to the double-stranded oligodeoxynucleotide sequence, allowing for targeted sequencing of off-target DSB locations.

DISCOVER-Seq offers an innovative live-cell approach by tracking MRE11 recruitment to DNA breaks via immunoprecipitation and sequencing, promoting real-time monitoring of Cas9-induced cleavage events. An advanced version, DISCOVER-Seq+ (ref. 143), increases sensitivity by inhibiting NHEJ pathways. Although in vitro methods can generate a high number of false positives, cell-based methods might miss off-target mutations occurring at frequencies below approximately 0.1% in nuclease-treated cell populations. Therefore, these approaches are complementary to each other.

Box 1 | Application of pairwise libraries for genome-wide functional screening

One of the major applications of gene-editing tools is genome-wide functional screening. Compared with zinc-finger nucleases (ZFNs) and transcription activator-like effector nucleases (TALENs), which require complex and labour-intensive protein engineering to target different genes, CRISPR–Cas nucleases and the genome-editing tools derived from them enable easier reprogramming of the target gene by simply changing the short guide RNA sequences. This flexibility enables large-scale perturbation screening, supporting systematic investigation of the functions of hundreds to thousands of genes in various contexts. For example, CRISPR-based screening has been applied to identify genes associated with drug resistance¹⁵, therapeutic targets in cancer^{17,178}, host factors involved in viral infection^{179,180} and genes crucial for T cell function¹⁸¹.

Following these advancements, base editors and single guide RNA (sgRNA) libraries were utilized to measure the effects of genetic variants in *BRCA1* and *BRCA2*¹⁷, and to evaluate cancer-related transition mutations^{18,19}. Prime editing was also adopted for multiplexed screening, providing the capability to introduce diverse genetic variants, including transversion mutations and small indels. In 2022, saturation prime editing for several regions in the *NPC1* and *BRCA1* genes was conducted using PE2 in haploidized cell lines²¹. In the same year, the functional effects of approximately 5,000 variants identified by genome-wide association studies or reported in the ClinVar database were investigated using PE3 in the MCF7 cell line²². Notably, introducing and evaluating over 1,000 genetic variants within

endogenous *TP53* demonstrated that the phenotypes resulting from some *TP53* variants differed from those observed using conventional exogenous overexpression systems, highlighting the potential of prime-editing screening to overcome the artefacts associated with the functional evaluation of variants using cDNA-based variant overexpression methods²³.

The major limitations of base editing include the variability in outcomes caused by sgRNAs, which can lead to the generation of bystander mutations around the target nucleotide or base-editor-induced indels, introducing noise into data analysis. In the case of prime editing, although the purity of editing outcomes is generally higher, the efficiency is often low, making it difficult to detect phenotypic changes. As a result, a screening approach using a pairwise library, rather than a library containing only sgRNA or prime editing guide RNA sequences, has been proposed^{18,19,23} (Fig. 3c). In this method, the corresponding integrated target sequences (often referred to as a 'surrogate target sequences', 'reporters' or 'sensors') were used to measure genome-editing outcomes and efficiencies, with this information being used to interpret screening results. In 2022, researchers tracked base-editing efficiencies and outcomes at surrogate target sequences and identified sgRNAs that produced a single predominant protein variant per sgRNA^{18,19}. For prime editing, surrogate target sequences called sensors were also harnessed to filter out inefficient prime editing guide RNAs, enabling more accurate functional evaluation of variants generated by prime editing²³.

Methods based on uracil excision and strand cleavage by USER enzymes, and inosine recognition and cleavage by EndoV^{144,145} have also been designed to identify genome-wide off-target sites for CBEs and ABEs, respectively. These methods have higher sensitivity in detecting off-target effects than Cas9-detection methods, enabling more precise characterization of off-target sites of base editors. For prime editors, methods such as nDigenome-seq¹⁰⁶, TAPE-seq¹⁰⁴ and PE-tag¹⁰⁵ have been developed; nDigenome-seq detects H840A nickase-induced single-strand breaks in vitro (the result of the initial step of prime editing) by comparing sequencing patterns in treated versus untreated controls. TAPE-seq and PE-tag are cell-based methods that introduce tag sequences for specific amplification and have enabled the identification of prime editor-induced off-target sites in both mammalian cell lines and mouse liver. However, nDigenome-seq does not capture subsequent processes after the nicking event, and both TAPE-seq and PE-tag introduce a tag into the pegRNA, which alters its sequence from the one used in the actual editing process. This sequence alteration could influence on-target and off-target prime-editing efficiencies. Despite these limitations, these off-target evaluation methods provide valuable insights into the specificity and safety of base and prime editors, complementing methods used for CRISPR nucleases. Building on these advances, Tracking-seq¹⁴⁶ has been introduced as a broadly applicable method for in situ off-target site detection. This approach involves capturing replication protein A-bound ssDNA and performing strand-specific library construction, which enables the detection of off-target sites that have been cleaved or nicked by DSB-inducing or single-strand break-inducing genome editors such as Cas9, base editors and prime editors in both in vitro and in vivo settings.

Machine learning applications

High-throughput profiling experiments for various genome-editing tools have enabled researchers to acquire extensive information about editing efficiencies and specificities across thousands to tens of thousands of target DNA sequences. Machine learning models trained on these datasets can facilitate the selection of efficient and specific sgRNAs and pegRNAs, circumventing the need for time-consuming and labour-intensive wet experiments (Fig. 3a). Initial efforts to develop machine learning models for predicting genome-editing efficiencies and specificities were based on medium-scale datasets (~4,400 sgRNAs) of phenotypic changes in gene-edited cells^{109,110,147–150}. However, these models often exhibited limited generalization performance, likely due to the inadequate size and quality of the training datasets.

Predicting Cas nuclease activity

Advancements in pairwise library-based approaches have generated large datasets of activities for Cas nucleases, base editors and prime editors, leading to the development of highly accurate machine learning models. Owing to the sufficiently large size of these datasets, which often include over 10,000 guide–target pairs, not only conventional machine learning models but also deep learning-based models that can autonomously identify important features from target DNA sequence information have been devised. For example, in 2017, a deep learning model named DeepCpf1 was developed to predict Cas12a (previously referred to as Cpf1)-induced indel frequencies. Indel frequencies at 15,000 target sequences obtained by pairwise library screening were used. Fine-tuning of this model using chromatin accessibility data and indel frequencies from about a hundred endogenous target sites further enhanced prediction accuracy¹⁵¹. A deep learning model

named DeepSpCas9 was also developed for SpCas9 using pairwise library screening data, which demonstrated higher generalization performance than previous prediction models¹¹⁷. Models developed

subsequently have substantially reduced the time and cost required to select the most suitable sgRNA for SpCas9 and its variants^{118–120,152} at given targets. Regarding RNA nucleases, a prediction model for

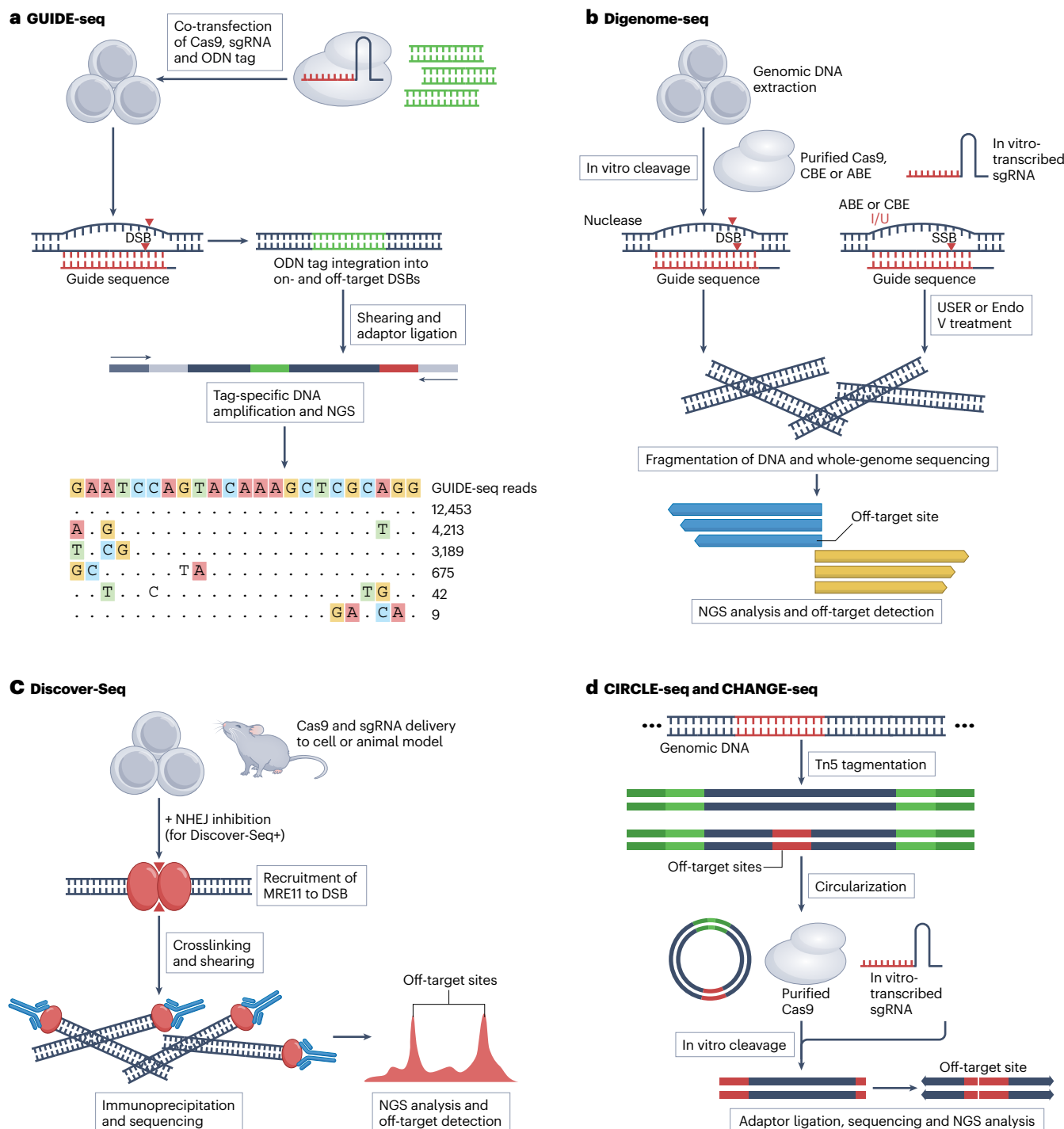


Fig. 4 | Schematics of unbiased, genome-wide off-target detection methods. **a**, GUIDE-seq detects off-target cleavage sites by integrating short double-stranded oligonucleotide (ODN) tags into the double-strand break (DSB) points. **b**, Digenome-seq identifies off-target sites by whole-genome sequencing of genomic DNA that has been cleaved in vitro by Cas nucleases. **c**, DISCOVER-Seq+ enables the

detection of off-target sites both in vitro and in vivo by tracking the recruitment of the DNA damage repair protein MRE11. **d**, CIRCLE-seq and CHANGE-seq measure Cas9-induced cleavages in a circularized genomic DNA library. ABE, adenine base editors; CBE, cytosine base editors; NGS, next-generation sequencing; NHEJ, non-homologous end joining; sgRNA, single guide RNA; SSB, single-strand break.

Cas13d from *Ruminococcus flavefaciens* XPD3002 (RfxCas13d) was developed using a dataset of the activities of approximately 24,000 crRNAs targeting a fluorescent protein and cell-surface marker¹⁵³. In 2024, a state-of-the-art model was developed using data from around 200,000 crRNAs targeting essential genes, providing on-target and off-target predictions that outperform existing models¹⁵⁴.

Predicting indel outcomes

Previously, indels generated during the repair of DSBs were thought to occur randomly. In 2016, analysis of DNA repair outcomes at 223 sites in the human genome demonstrated that indel patterns are reproducibly consistent and follow a non-random distribution across experimental replicates¹⁵⁵. Building on this finding, prediction of editing outcomes generated by Cas9 nucleases^{115,116} enabled precise, donor DNA template-free genome editing of mutations associated with genetic diseases. By analysing large-scale datasets of edited targets containing indels via pairwise library screening, the effects of the sequence context around the cut site on indel outcomes during repair of DSBs was identified, leading to the development of models such as inDelphi and FORECast, which demonstrate high accuracy in predicting indel patterns in diverse cell types. Using predictions from these models, causative mutations of Hermansky–Pudlak syndrome and Menkes disease in patient-derived cells were corrected by delivering only Cas9 and sgRNAs¹¹⁵.

Predicting base-editor activity

Prediction models for base-editing efficiencies and outcomes have also been developed to facilitate the installation or correction of disease-relevant mutations^{122–127}. These models, capable of predicting base-editing efficiencies and outcomes across target sequences, were generated using data from tens of thousands of sgRNAs and corresponding target sites. These tools can predict not only the base-editing efficiency but also the proportions of multiple editing outcomes within the editing window. Additionally, the model BE-HIVE¹²² can predict non-C-to-T and non-A-to-G conversions for CBEs and ABEs. Adopting Cas9 variants with different PAM compatibilities and combining them with appropriate base-converting domains (that is, the deaminase domain) can optimize the editing position within the base-editing window, thereby maximizing the intended editing efficiency and minimizing bystander effects. However, measuring the efficiencies of these variants is extremely time-consuming and costly. To address this issue, base-editor efficiencies and outcomes for 63 base editors (nine Cas9 variants with different PAM compatibilities times seven deaminase domains) were predicted at a given target sequence using fine-tuning of deep learning models¹²⁶. When base editor–sgRNA combinations were selected using these models, the predicted efficiencies increased 2.9-fold to 20-fold with respect to those of rationally designed SpCas9-containing base editor–sgRNA combinations.

Predicting prime-editor activity

Designing optimal pegRNAs is highly challenging as it requires consideration of not only the guide sequence but also the composition and lengths of the PBS and RTT sequences. This issue increases the number of possible design options for pegRNAs compared with sgRNAs by tens to hundreds of times. Analysing the effects of PBS and RTT lengths on prime-editing efficiencies using a library of 54,836 pegRNA–target sequence pairs led to the development of prototype prediction models DeepPE, PE-type and PE-position¹²⁹. However, these tools were limited by the types and positions of the edits, and the lengths of the PBS and RTT,

primarily due to the insufficient size of the training datasets. In 2023, MinsePIE¹³⁰, which predicts prime-editor-induced insertion rates, and PRIDICT¹³¹, which focuses on 1 bp replacements as well as short insertions and deletions, were developed using data from 3,604 and 92,423 pegRNAs, respectively. Following these advancements, DeepPrime¹²⁸, a state-of-the-art model trained on data from 338,996 pegRNAs, was developed, enabling the prediction of prime-editing efficiencies for eight prime-editing systems across seven cell types for all possible types of 1 bp to 3 bp edits at target sites with both NGG and non-NGG PAMs. More recently, the prediction scope of PRIDICT was greatly extended by using an expanded dataset of the activities of over 400,000 pegRNAs to cover edits of up to 15 bp in length (PRIDICT2.0) and incorporating the effects of local chromatin environments on prime-editing efficiencies for target sites with NGG PAMs (ePRIDICT)¹³². These prediction models should facilitate functional variant mapping using prime-editor-mediated saturation genome editing¹³⁴ and the development of highly efficient gene-editing therapeutics⁹⁹. For instance, prime editor–pegRNA combinations to introduce genetic variants in NPC1 and BRCA2 were designed using DeepPrime, resulting in efficiencies 2.1-fold to 7.7-fold higher than those obtained with rational design^{21,128}.

Predicting off-target effects of genome-editing tools

A major safety concern in translational research involving genome-editing tools is undesired off-target effects. Unbiased off-target detection methods (Fig. 4), combined with high-throughput profiling methods described in the previous section, have facilitated the development of machine learning models to predict guide sequence-dependent, Cas9-induced off-target effects^{110,141,156,157}. In addition, pairwise libraries containing mismatched target sequences have been used for training off-target prediction models for base editors¹²⁵ and prime editors¹²⁸. Despite advancements in predictive models for selecting sgRNAs and pegRNAs to address off-target issues, several challenges remain unresolved. First, limited information is available about the sequence composition conditions under which large, Cas9-induced deletions occur at on-target sites, making it nearly impossible to design sgRNAs that would avoid this issue. Second, the guide RNA-independent off-target effects of base editors are almost stochastic, although there are some motifs that appear to be preferred by the specific deaminases, such as ACW (in which W = A or T) by rAPOBEC1 and UA by TadA⁸⁴. Such off-target effects could occur genome-wide or transcriptome-wide, making them challenging to consider in sgRNA design. Therefore, researchers require time-consuming and labour-intensive wet-lab experiments to evaluate these off-target issues.

AI-based development and evolution of genome-editing tools

Recent advances in biological language models trained on biological data have enabled the generative design and machine-guided evolution of genome-editor proteins, offering an alternative to conventional approaches^{158,159}. In 2024, a genomic foundation deep learning model named Evo¹⁵⁹, trained on millions of prokaryotic and phage genomes, was able to design synthetic CRISPR systems de novo. Fine-tuning this foundational model with a dataset of 72,831 CRISPR–Cas loci enabled the generative design of Cas9–sgRNA complexes. Notably, although the generated Cas9 protein and sgRNA exhibited only 73.1% and 91.1% sequence identity with those of SpCas9, respectively, they demonstrated robust in vitro cleavage activity, highlighting the potential of AI-driven de novo design to generate functional genome-editing tools with minimal similarity to natural counterparts.

For example, a protein language model-based framework, EVOLVEpro¹⁵⁸, leverages machine learning-guided protein evolution, requiring only a few iterations to optimize protein activity. EVOLVEpro integrates a protein language model, ESM-2 15B (ref. 160), with a regression model in a modular framework to predict protein activity landscapes. In this process, EVOLVEpro generates variants in silico, which are then validated through wet-lab experiments. The experimental results are fed back into the model, enabling iterative improvements in activity predictions. Compared with conventional directed evolution methods, this few-shot training approach minimizes the experimental workload by requiring only a small number of data points to optimize protein activity. For instance, the compact Cas12f nuclease from *Pseudomonas aeruginosa* (PsaCas12f) and the prime editor M-MLV RT domain were optimized after four rounds of evolution, with 12 single mutations per round, achieving 2.2-fold to 44-fold and 1.4-fold to 1.8-fold increases in editing efficiency, respectively.

To summarize, high-throughput evaluation methods have not only facilitated the experimental validation of genome-editing tools but also enabled the development of machine learning models for sgRNA and pegRNA design. These models enable researchers to streamline wet-lab experimental designs and address the practical limitations of high-throughput experiments. Moreover, the cumulative data generated from high-throughput methods can complement generative AI technologies, greatly facilitating the development of highly efficient and precise genome-editing tools.

Outlook

Despite the potential of one-shot gene-editing therapy, the development of CRISPR nucleases as therapeutics often remains confined to specific diseases that can be treated by the NHEJ-mediated knockout of relevant genes. Base and prime editors have expanded the scope of in vivo gene therapeutics, making it possible to target about 90% of known pathogenic mutations. Furthermore, they alleviate safety concerns associated with gene therapy because they almost entirely avoid issues such as DSB-induced cell death, large on-target deletions and off-target indel generation. Both base editors and prime editors have been validated in animal models to achieve ~63% and ~46% gene-editing efficiencies, respectively, in mouse liver^{99,102,161–165}.

New genome-editing strategies are continuously being proposed to broaden the scope of gene editing and enhance editing efficiency. Notable advancements include improvements in prime-editing efficiency, enabling the introduction of large insertions or deletions spanning several kilobases using two pegRNAs that bind to opposite DNA strands^{166–170}. Additionally, technologies such as PASTE (programmable addition via site-specific targeting elements)¹⁷¹ and PASSIGE (prime-editing-assisted site-specific integrase gene editing)^{167,172} facilitate targeted large gene insertions by using prime editors to install large serine recombinase attachment sites and serine integrases simultaneously. These technologies could greatly contribute to the treatment of genetic diseases caused by various loss-of-function mutations within specific genes through the introduction of functional genes.

As the number and complexity of gene-editing components increases (for example, two pegRNAs, one pegRNA and one sgRNA, or two sgRNAs, along with their corresponding target sequences), evaluating these genome-editing technologies using library approaches becomes increasingly challenging. Current array-based oligonucleotide library synthesis technologies are generally limited to lengths of 300–400 nucleotides, making it difficult to synthesize multiple editing components simultaneously. Furthermore, precise library cloning and

analysis methods are hindered by issues such as the uncoupling of DNA components due to template switching during PCR¹³⁵ and/or lentiviral transduction¹³⁶. To overcome such limitations, methods for connecting multiple oligo pools have been proposed¹⁷³, and experimental techniques are being optimized to reduce template switching^{174–176}. These advancements could facilitate the high-throughput profiling of more complex genome-editing tools and technologies.

Genome-wide CRISPR screening is widely used to induce frame-shift-mediated knockout of target genes to identify genes related to drug resistance, profile essential genes or determine host factors associated with pathogen infection. Furthermore, screening technologies have been developed to introduce genetic variants en masse using HDR, base editing and prime editing to elucidate the functional effects of numerous discovered genetic variants. Prime editing is particularly promising for genome-wide screening as it can overcome the limitations of HDR, which has very low efficiency in mammalian cells, necessitating the use of haploid cells and base editing, which only allows for certain base conversions. Although current technology focuses on specific genes due to the insufficient efficiency of prime editing for investigating variants across multiple genes, continuous advancements in prime-editing technology, high-throughput evaluation for activity optimization and the use of machine learning models will enable accurate functional screening of genetic variants in the future.

High-throughput methods that enable direct comparisons between Cas nucleases, base editors and prime editors at thousands of target sequences would help identify the most effective genome-editing tool for introducing the desired edit at a given target site. Massively parallel evaluations of various Cas9 variants¹¹⁸ and orthologues¹²¹ have promoted the selection of the most efficient and safe variant or orthologue for specific target sites. Moreover, by considering diverse combinations of nickases and deaminases, the optimal base editor for specific target nucleotide conversions can be identified^{122,126}. For prime editing, the most suitable prime-editor system can be chosen according to the intended edit and the DNA repair systems in specific cell types¹²⁸. Currently, comparisons are made separately among given types of genome-editing tools. Direct comparison would instead enable the selection of the optimal tool and corresponding sgRNA/pegRNA for each experimental situation, improving both basic and translational research in gene editing.

The continuous advancements in CRISPR-based genome-editing technologies are transforming the landscape of genetic research and therapeutic development. High-throughput evaluation-based optimization of base editors, prime editors and novel genome-editing tools will facilitate their applications.

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H.H.K. conceptualized the manuscript. H.H.K. and H.H.K. wrote the manuscript.

Competing interests

H.H.K. is the founder of cisionMed. H.H.K. declares no competing interests.

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